Name
Date
Course

Sequences and Series (Part III) Homework

Find each sum. Do not use a formula.

① $\sum_{n=1}^{4} -5n-2$

② $\sum_{n=0}^{4} n^2+9$

③ $\sum_{n=2}^{5} 6n$

Warning! $-4^{n-3} \neq (-4)^{n-3}$

④ $\sum_{n=3}^{5} -4^{n-3}$

Determine if each series is arithmetic or geometric. Then find the sum. Use formulas.

⑤ $\sum_{n=1}^{19} 9+3n$

⑥ $\sum_{n=1}^{44} 1000\left(-\frac{1}{2}\right)^n$

⑦ $\sum_{n=1}^{15} -\frac{4}{5}n$

⑧ $\sum_{n=1}^{66} (0.4)^n$

(9) $\sum_{n=2}^{12} 5^{n-1}$

(10) $\sum_{n=0}^{37} \frac{n}{2}-1$

11 $\sum_{n=4}^{90} 6+n$

12 $\sum_{n=3}^{9} (1.8)^{n-2}$

(13) $\sum_{n=1}^{16} 4n-4$

(14) $\sum_{n=1}^{21} \frac{n-3}{2}+1$

(15) $\sum_{n=2}^{39} -10n+20$

(16) $\sum_{n=3}^{11} 81-27n$

(17) $\sum_{n=0}^{56} 7n$

Find the value of each infinite geometric series if it exists.

(18) $\sum_{n=1}^{\infty} (-0.3)^n$

★ (19) $\sum_{n=2}^{\infty} \left(-\frac{3}{2}\right)^{n-1}$

(20) $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$

☆ (21) $\sum_{n=3}^{\infty} (2.3)^{n+1}$

Warning! $-\left(\frac{3}{5}\right)^n \neq \left(-\frac{3}{5}\right)^n$

(22) $\sum_{n=2}^{\infty} -\left(\frac{3}{5}\right)^n$

Answers

(1) $\boxed{-58}$

(2) $\boxed{75}$

(3) $\boxed{84}$

(4) $\boxed{-21}$

(5) $\boxed{\text{Arithmetic}}$ $\boxed{741}$

(6) $\boxed{\text{Geometric}}$

$\boxed{-\frac{1000}{3}\left(1-\left(-\frac{1}{2}\right)^{44}\right)}$

or $\boxed{-\frac{1000}{3}\left(1-\left(\frac{1}{2}\right)^{44}\right)}$

or $\boxed{\frac{1000}{3}\left(\left(\frac{1}{2}\right)^{44}-1\right)}$

(7) $\boxed{\text{Arithmetic}}$ $\boxed{-96}$

(8) $\boxed{\text{Geometric}}$ $\boxed{\frac{2}{3}(1-0.4^{66})}$

(9) $\boxed{\text{Geometric}}$ $\boxed{-\frac{5}{4}(1-5^{11})}$

(10) $\boxed{\text{Arithmetic}}$ $\boxed{313.5}$

(11) $\boxed{\text{Arithmetic}}$ $\boxed{4,611}$

(12) $\boxed{\text{Geometric}}$ $\boxed{-\frac{9}{4}(1-1.8^7)}$

(13) $\boxed{\text{Arithmetic}}$ $\boxed{480}$

(14) $\boxed{\text{Arithmetic}}$ $\boxed{105}$

(15) $\boxed{\text{Arithmetic}}$ $\boxed{-7,030}$

(16) $\boxed{\text{Arithmetic}}$ $\boxed{-972}$

(17) $\boxed{\text{Arithmetic}}$ $\boxed{11,172}$

(18) $\boxed{-\frac{3}{13}}$

(19) Because $r=-\frac{3}{2}$ is not between -1 and 1, the sum does not exist.

(20) $\boxed{\frac{4}{3}}$

(21) Because $r=2.3$ is not between -1 and 1, the sum does not exist.

(22) $\boxed{-\frac{9}{10}}$